

Your grade: 100%

Your latest: 100% • Your highest: 100% • To pass you need at least 80%. We keep your highest score.

Next item →

1. Which algorithm generally has a lower time complexity for searching an item in a sorted list?

1 / 1 point

- ☐ Bubble search
- ☒ Binary search
- ☐ Sequential search
- ☐ Insertion search

👉 Nice work

Correct! Binary search operates in

$$O(\log n)$$

time complexity, making it more efficient for sorted lists compared to sequential search.

2. What is a complete binary tree?

1 / 1 point

- ☒ A binary tree in which all levels are fully filled except possibly the last, which is filled from left to right.
- ☐ A binary tree with nodes only on the leftmost side.
- ☐ A binary tree in which all nodes have left and right children of the same height.
- ☐ A binary tree in which each node has exactly two children, except for the leaves.

👉 Nice work

Correct! A complete binary tree fills all levels, except possibly the last, which is filled from left to right.

3. Which of the following are key components when writing pseudocode to represent an algorithm?

1 / 1 point

- ☐ Include error handling for unexpected inputs.
- ☒ Initialize variables to store data.

👉 Nice work

Correct! Initializing variables is essential for storing and manipulating data in algorithms.

- ☒ Break down the problem into smaller subproblems.

👉 Nice work

Yes, breaking down problems helps in detailing the algorithm's approach in pseudocode.

- ☒ Outline the steps of the algorithm in logical order.

👉 Nice work

Well done! Outlining steps logically is critical to effectively communicate the algorithm's process.

- ☐ Use specific programming language syntax.

4. What is one of the key benefits of using the divide and conquer algorithm design technique?

1 / 1 point

- ☐ It guarantees the optimal solution for all problems.
- ☒ It significantly reduces the problem size at each step.
- ☐ It eliminates the need for recursion in algorithm design.
- ☐ It only uses iterative loops to solve problems.

👉 Nice work

Great job! The divide and conquer technique effectively reduces the problem size by breaking it down into smaller subproblems.

5. What is the primary advantage of using the insertion sort algorithm over bubble sort in the context of sorting small datasets?

1 / 1 point

- ☐ Insertion sort is more efficient for sorting datasets with repeated elements.
- ☒ Insertion sort has a better average-case time complexity.
- ☐ Insertion sort requires less memory overhead than bubble sort.
- ☐ Insertion sort can handle larger datasets more efficiently than bubble sort.

👉 Nice work

Correct! The average-case time complexity of insertion sort is better suited for small datasets compared to bubble sort.